

STUDENT	
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Student number	s3693414
Faculty	Faculty of Electrical Engineering, Mathematics and Computer Science
Research chair/group	Pervasive Systems
Study field	(MSc) Computer Science
Start date assignment	20/02/2026
End date assignment	13/04/2026

ASSIGNMENT COMMITTEE	
1. Committee Chair	Le Viet Duc, Duc, dr.
2. Committee member UT	
3. Committee member UT	
4. Additional member(s)	, ,

ASSIGNMENT DESCRIPTION	
Title of the Thesis	Machine Learning-Based Threat Detection for ISP Network Traffic
Assignment description	The research will first validate previous results reporting 95% accuracy and establish a baseline. It will then examine how contextual features and advanced machine learning methods impact both accuracy and real-time feasibility. Additionally, explainable AI techniques will be applied to improve model interpretability. The goal is to design a detection approach that is accurate, scalable, and transparent.
Additional Remarks	

COURSE CODE(S)	
Course Code	201800524
Second Course Code	
Third Course Code	
Fourth Course Code	

SIGNATURE	
Signature student University of Twente Name: Xiaoyan Xiong	Signature Chairman University of Twente Name: Le Viet Duc, Duc, dr.

20/02/2026

(date (dd-mm-yyyy))

Xiaoyan Xiong

(Signature)

20/02/2026

(date (dd-mm-yyyy))



(Signature)